Part 5

You are going to read a newspaper article about food. For questions **31–36**, choose the answer (**A**, **B**, **C** or **D**) which you think fits best according to the text. Mark your answers **on the separate answer sheet.**

The food scientist

Alice Baines meets the influential food scientist Charles Spence

Charles Spence will eat just about anything. 'We've got ice cream made from bee larvae at home,' says the Professor of Experimental Psychology in his office at Oxford
line 4 University in the UK. They may be maggoty in appearance, but they apparently have a 'slightly nutty, floral' flavour. How to make bug-eating acceptable is just one of the
line 7 many gustatory challenges that Spence and his team are tackling. Through his studies into how the senses interact to form our perception of flavour, Spence is influencing,
line 10 in a stealthy but not inconsiderable manner, what we eat
line 11 and drink, from the hefty output of food-industry giants (he sits on the scientific advisory board of one well-known multinational conglomerate, and receives funding from another), to the menus of leading restaurants.

Spence and his peers have, through a line of scientific inquiry that is informally referred to as gastro physics, studied in minute detail how we experience food and drink. Who we eat with; how food is arranged and described; the colour, texture and weight of plates and cutlery; background noise – all these things affect taste. Spence's book, *The Perfect Meal*, written with Betina Piqueras-Fiszman, is packed with insights that are fascinating to anyone in possession of an appetite.

Were you aware that the person in a group who orders first in a restaurant enjoys their food most? And did you realise that we consume about 35% more food when eating with one other person, rising to 75% more when dining with three others?

Spence's lab in Oxford is noticeably un-space-age. 'Low-tech, paper and drawing pin stuff,' he readily admits. There are soundproof booths that resemble human-sized safes ('most of my PhD was done in one of those,' he says fondly), along with stacks of ancient-looking audio-visual equipment. By keeping overheads low, he can afford to work more creatively with cooks who can't fund academic research themselves. Much of his work is funded by a major food multinational. Historically, he says, undertaking research which attracts industry funding has been seen in university circles as 'what you do if you can't do proper science'. But since the government insisted that universities demonstrate their work has an impact and that people are interested in it, this type of research has become a strategically good thing to do.

Spence is currently helping famous brands through (often government-imposed) reductions in salt and sugar. It is in their interests, he points out, to help loyal customers stay alive for longer. Perhaps surprisingly, many have been making these reductions furtively, behind closed doors. They do it gradually, so regular consumers don't notice the difference from one pack to the next. 'The research shows that when you tell people what you're doing, it makes them focus on the taste and for whatever reason they don't like it as much,' he says.

It was while working on a project for a major food producer that Spence was first introduced to Heston Blumenthal, the renowned experimental chef. 'At the time, people thought: "Science and food – that's horrible," although most food is scientific, in fact. Who better to change the mindset than Heston?' It was through Blumenthal's collaborations with Spence, who had been studying the effects of sound on flavour, that the 'Sound of the Sea' dish came about in Blumenthal's five-star restaurant. Interestingly, Spence says that members of the early-twentieth-century art movement, the Italian futurists, were 'doing sounds of croaking frogs with frogs' legs a century ago', but that it didn't catch on.

Now the food industry is applying Spence's sensory science to products left, right and centre. This includes his recent findings that higher-pitched music enhances sweetness, and lower-pitched and brassy sounds make food taste bitter. 'It's always surprising when shapes affect taste, or when a tune can impact on how you perceive a flavour,' he says. An airline will soon be matching music with food served to passengers. And last year, a well-known brand released a smartphone app that plays a concerto while your ice cream softens; they omitted to match the music to the taste, though, which is all too frequently the case according to Spence.

What, one wonders, are dinner parties like in the Spence household? There was the time they ate rabbit, with the fur wrapped around the cutlery. And the one at which they played with remote-controlled, multi-coloured light bulbs. 'We've had dinner parties with a tone generator, headphones and ten different drinks lined up to see whether they have different pitches.' Home, sweet shops, food conventions, international gastronomy conferences: they're all extensions of the lab to Spence.